

## Project Name

AROHAN: Autonomous Campus Safety System  
Detect. Verify. Respond.

## Team Name

Techeez

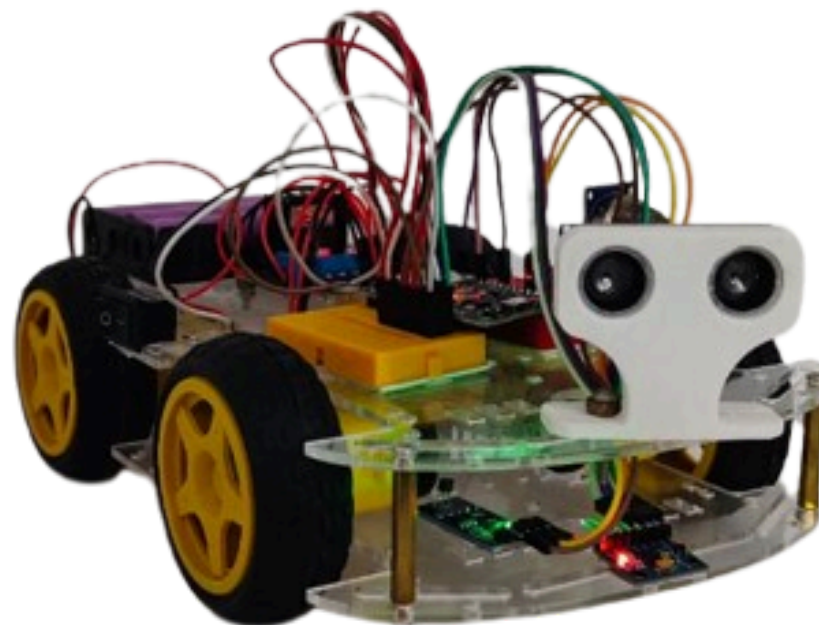
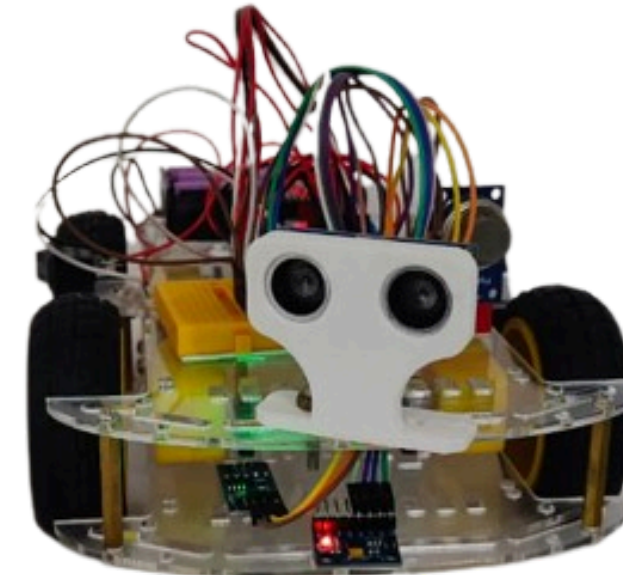
## Team Members

- Md Shadan Siddiqui
- Shivam Kushwaha
- Isha Kumari

## Institution

United College of Engineering and Research Prayagraj

**Contact** +91 7007055725  
shadanmd566@gmail.com



**FIXED IOT**

# THE PROBLEM WE ARE SOLVING

Across campuses, labs, industries, and workplaces, hazards like gas leaks, fires, chemical spills, and unauthorized access are **detected too late**, verified manually, and responded to slowly.

Current safety systems are isolated, reactive, and dependent on humans, causing delays of **2–5 minutes** that can turn a small fault into a catastrophe.

## WHY THIS MATTERS?

**80%**

Campus safety incidents (fires, gas leaks, intrusions) are caused by delayed detection & manual verification.

**2–5 min**

Average human verification delay before action is taken.

**60%**

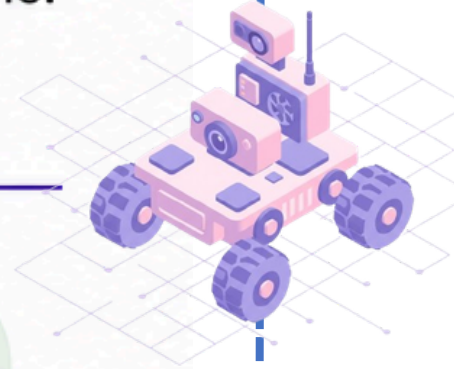
Higher chance of incident escalation when response takes > 2 minutes.

**Fragmented Systems**

Sensors, alarms, and cameras work in isolation and do not communicate.

**High Cost of Incidents**

Loss of lives, infrastructure, learning time, and environment.



## BHOPAL GAS TRAGEDY

📅 2–3 December 1984



Toxic gas leak from Union Carbide plant, Bhopal, India.



~15,000 – 20,000 lives lost instantly and in the following days.



Delayed detection, no real-time monitoring, and manual response led to disaster.

## VIZAG GAS LEAK TRAGEDY

📅 7 May 2020



Styrene gas leak from LG Polymers plant, Visakhapatnam, India.



12 lives lost, 1000+ people affected.



Leak went undetected for crucial minutes; no automated isolation or early warning.

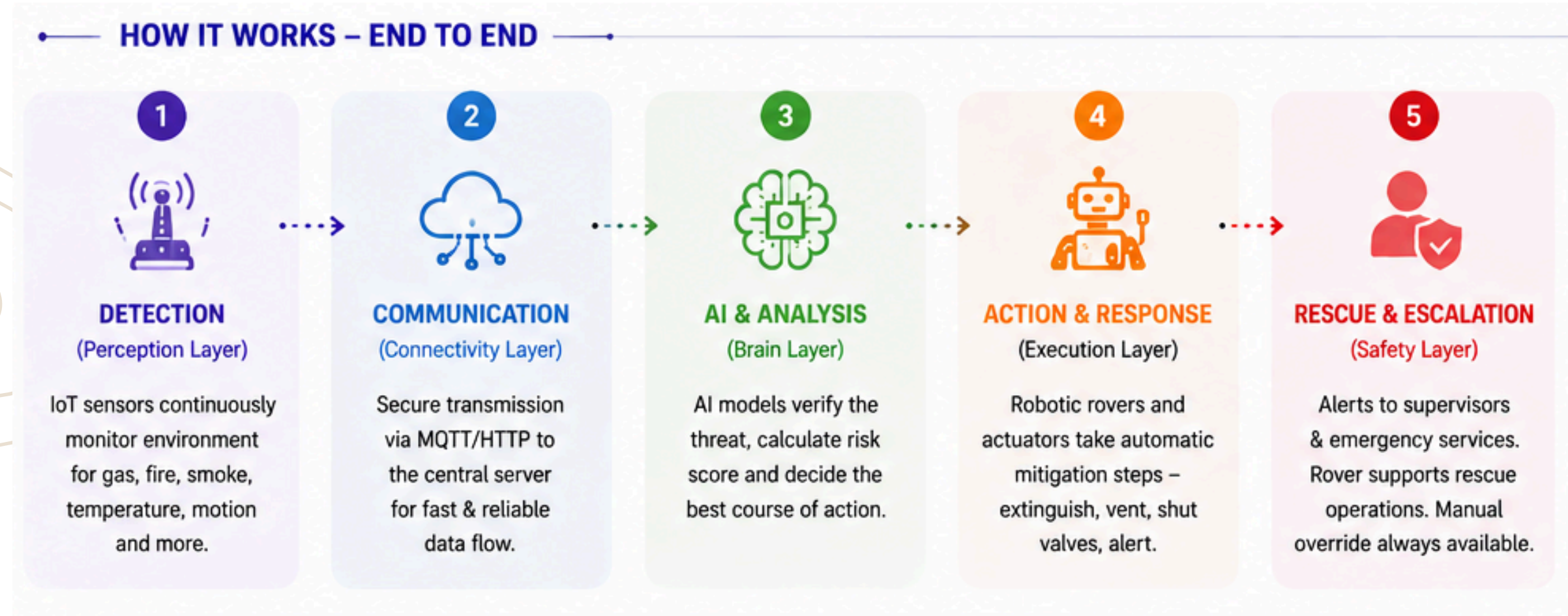


**THE CORE PROBLEM:** Safety systems only detect hazards but do not verify or act in time.

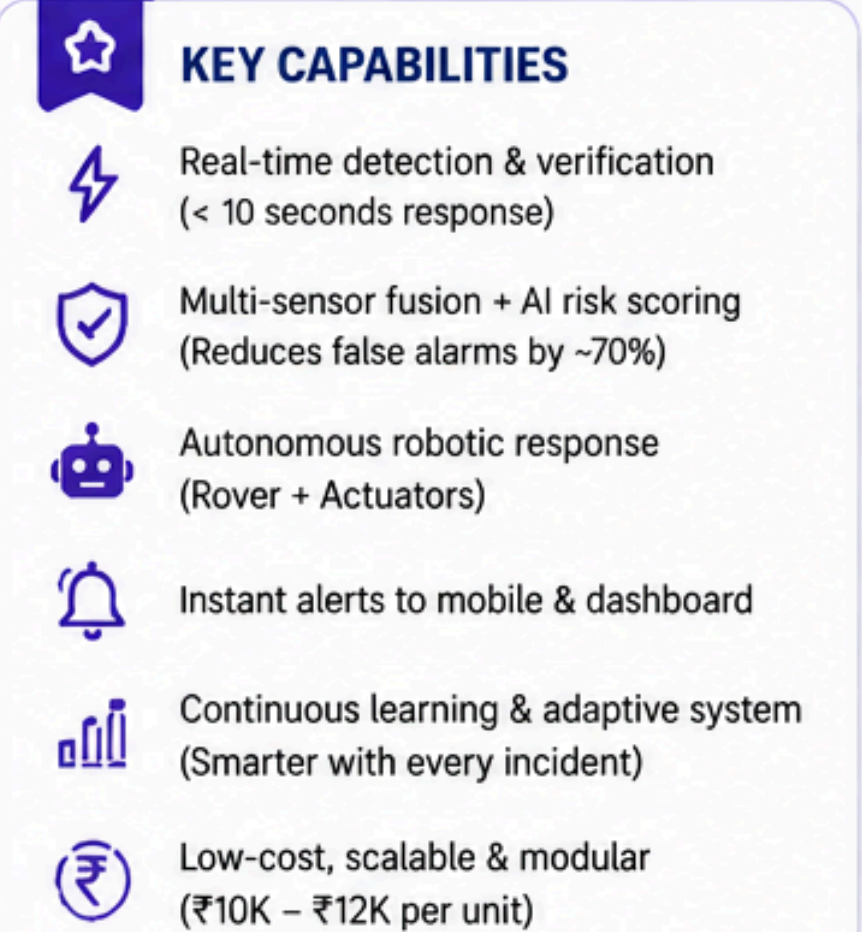
**We need an intelligent, autonomous system that can DETECT – VERIFY – RESPOND in real time.**

# Proposed Solution

## AROHAN- An AI powered Autonomous safety & response ecosystem

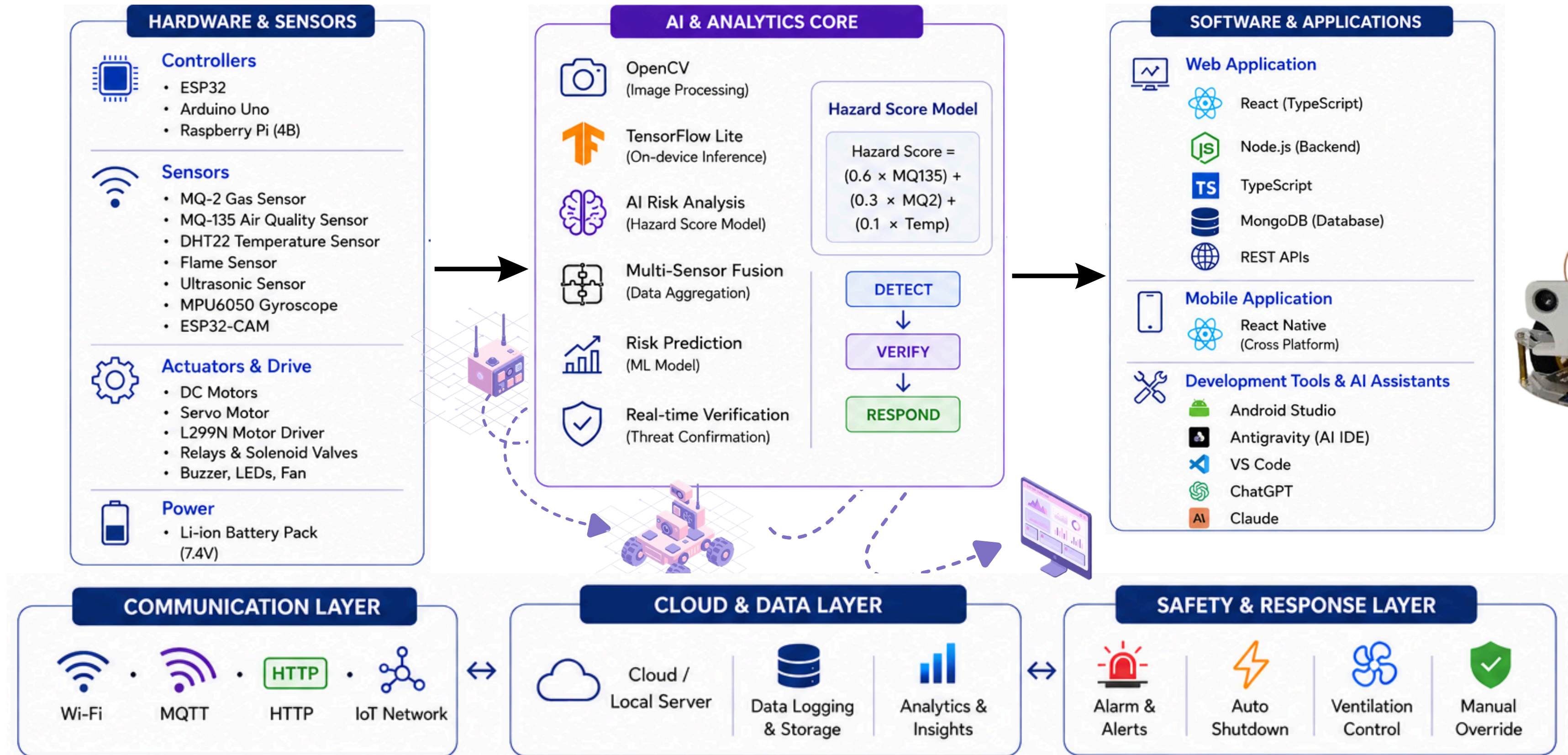


**DETECT (FIXED IOT) → VERIFY (AUTONOMOUS ROVER) → RESPOND (ALERT & MITIGATION)**



# Technology Stack

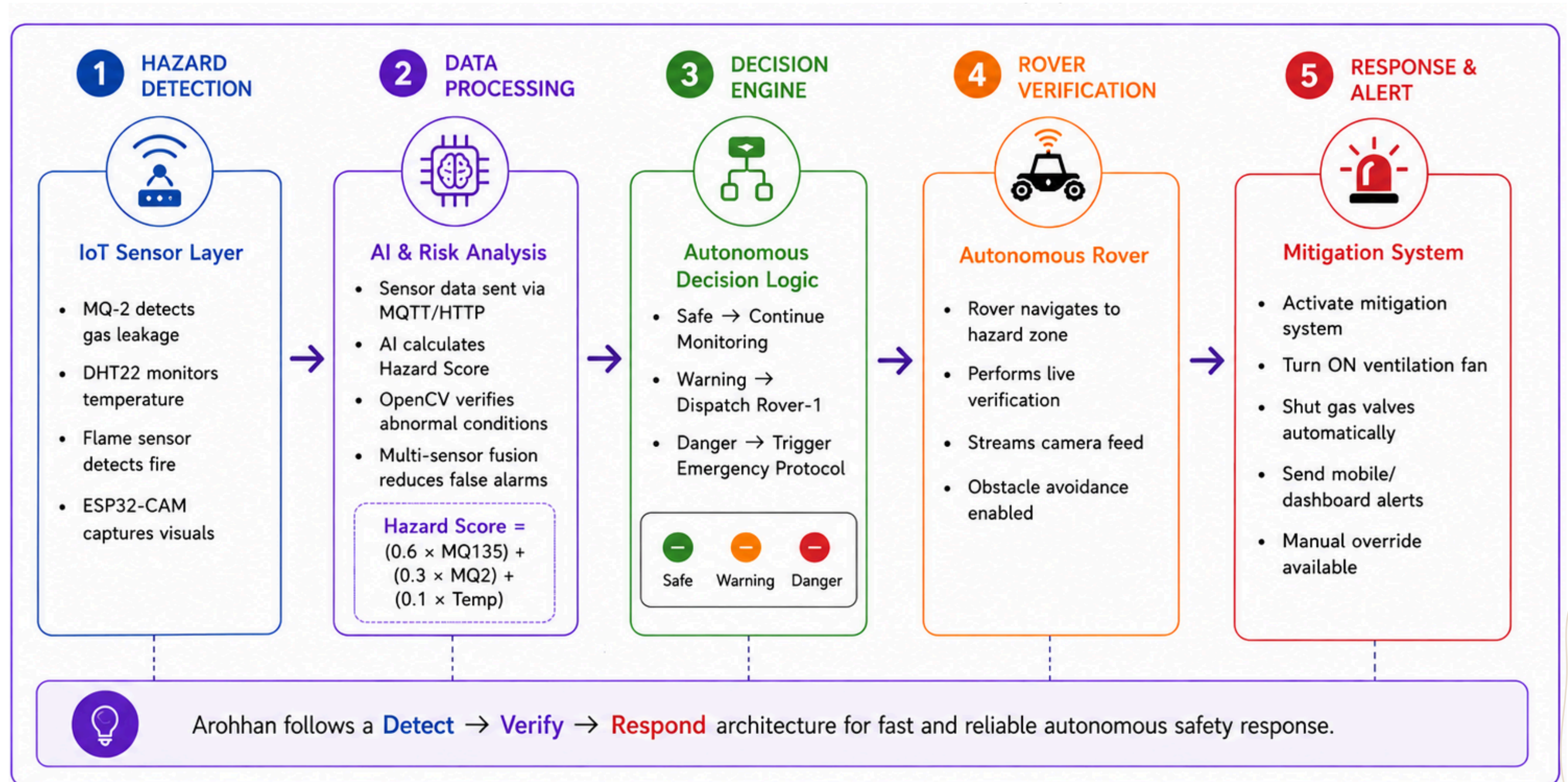
THE TOOLS, FRAMEWORKS AND TECHNOLOGIES POWERING AROHAN



Demo video link- [https://drive.google.com/file/d/1l-GsaEnuhVUpobU\\_Cl98upZ1kNSIQ63/view?usp=sharing](https://drive.google.com/file/d/1l-GsaEnuhVUpobU_Cl98upZ1kNSIQ63/view?usp=sharing)

# Implementation Plan

END TO END WROKFLOW OF AROHAN AUTONOMUS SAFETY SYSTEM



# Impact & Feasibility



## 1. REAL-WORLD IMPACT



- Reduces delayed hazard response
- Prevents gas leaks & lab fires
- Enables autonomous safety monitoring
- Reduces human dependency



## 2. WHO BENEFITS?



Students & Faculty



Laboratories & Campuses



Industries & Factories



Security & Maintenance Teams



## 3. FEASIBILITY



Low-cost prototype (~₹10–12K)



Uses affordable ESP32 & sensors



Scalable modular architecture



Easy real-world deployment



## 4. EXPECTED CHANGE



Faster emergency response (<10 sec)



Safer campuses & industries



Reduced false alarms



Smart autonomous safety ecosystem



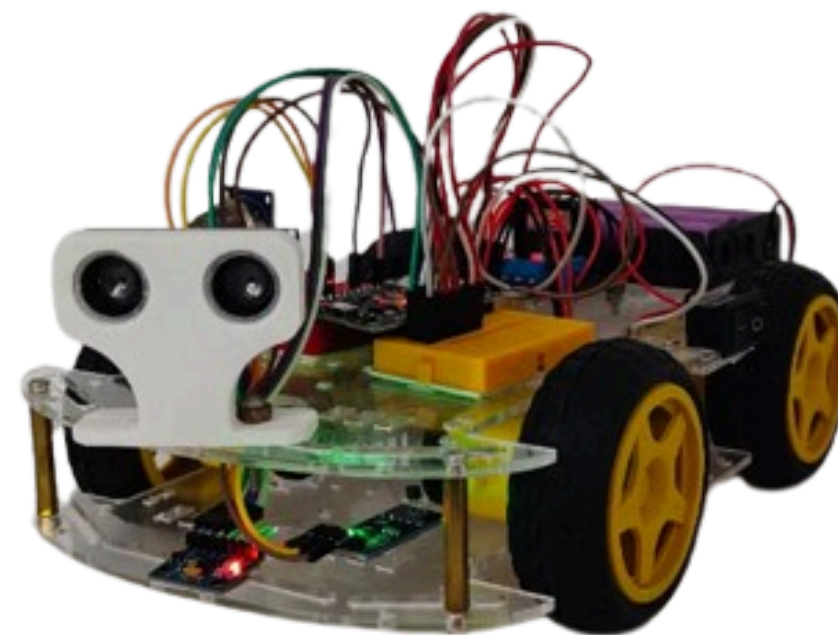
Arorhan transforms passive safety systems into an **intelligent autonomous response ecosystem.**

# Demo Video & MVP

## Links

1. Demo Video Link- [https://drive.google.com/file/d/1l-GsaEnuhVUpobU\\_\\_CI98upZ1kNSIQ63/view?usp=sharing](https://drive.google.com/file/d/1l-GsaEnuhVUpobU__CI98upZ1kNSIQ63/view?usp=sharing)

2. MVP Link- <https://arohan-ruby.vercel.app/>



# THANK YOU